complaint-router

Routing consumer financial complaints to the right team, with a system that keeps learning

KSE MLOps · Lab 1 · Zakhar Staryk · github.com/goodfrog33/kse-mlops-complaint-router

18.5 M
complaints in the CFPB database

3.8 M
with complaint text

10
product queues

At a glance

Problem

- ~21,000 complaints a day to the US regulator CFPB
- Each must reach the team that owns the product

System

- Predicts 1 of 10 product queues from the complaint text
- Confident → auto-routed; otherwise → human
- Retrained on drift, monitored before labels arrive

Data

- CFPB Consumer Complaint Database (public)
- 3.8 M complaint texts, 2015–2025
- Replayed month by month

Fit with the 5 labs

- Yes: pipeline, LLM, serving and agent all have real material
- Caveat: no new complaint text after 2025

Every complaint has to reach the team that owns it



~21,000 / day

complaints in 2026 (measured)

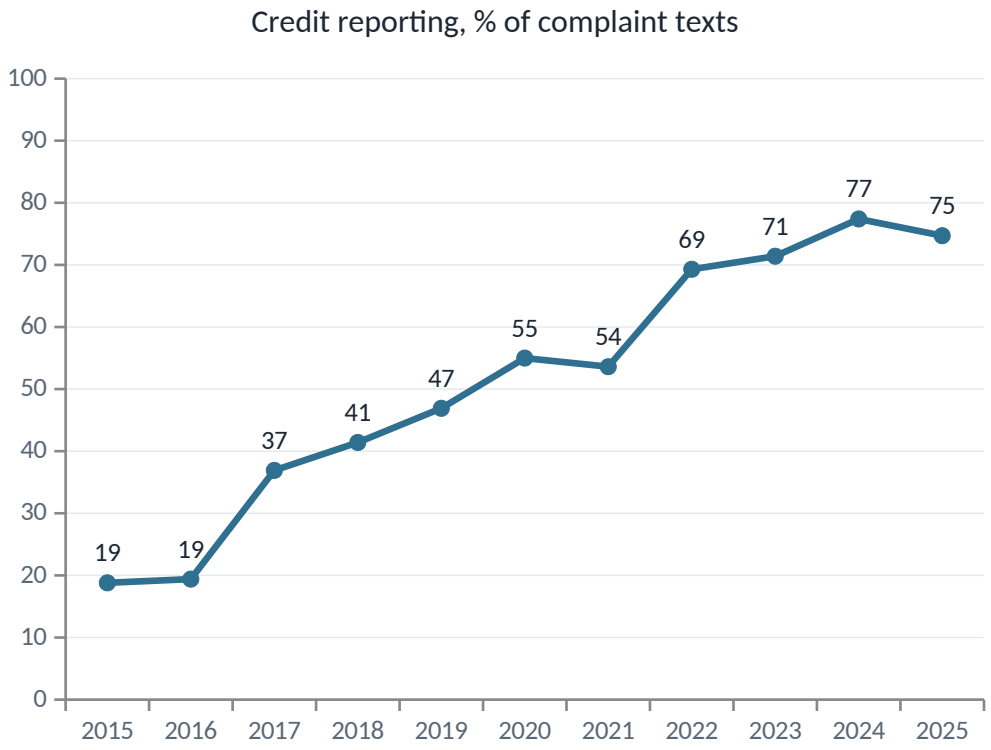
15 days

response window per complaint

Systematic

errors hide until labels return

The data keeps moving, so a one-off notebook would decay



2 taxonomy changes

2017-04 and 2023-08

19% → 77%

credit reporting share

30× volume

0.17 M → 5.4 M a year

Text stopped

CFPB, 2026

Success is measured on three separate layers

Business

≤ 8%
routed to the wrong queue

≥ 60%
routed without a human

Model

≥ 0.80
macro-F1

≥ 0.95
precision on auto-routed

System

≤ 300 ms
p95 latency

≥ 99.5%
availability

≤ 3 days
data freshness

Every non-functional requirement carries a number

Category	Requirement
Freshness	data ≤ 3 days · labels after 15 days
Latency	p95 ≤ 300 ms · 20 req/s per replica (need ~2.5)
Availability	≥ 99.5% · fallback to human queue
Reproducibility	model = commit + DVC version + params
Adaptability	monthly and alert-triggered retraining
No-label observability	input, prediction, confidence drift
Resources, cost	4 GB Minikube · ≤ $800/month in cloud

Conflict

Freshness vs reproducibility

Reproducibility wins

Conflict

Precision vs coverage

Precision wins

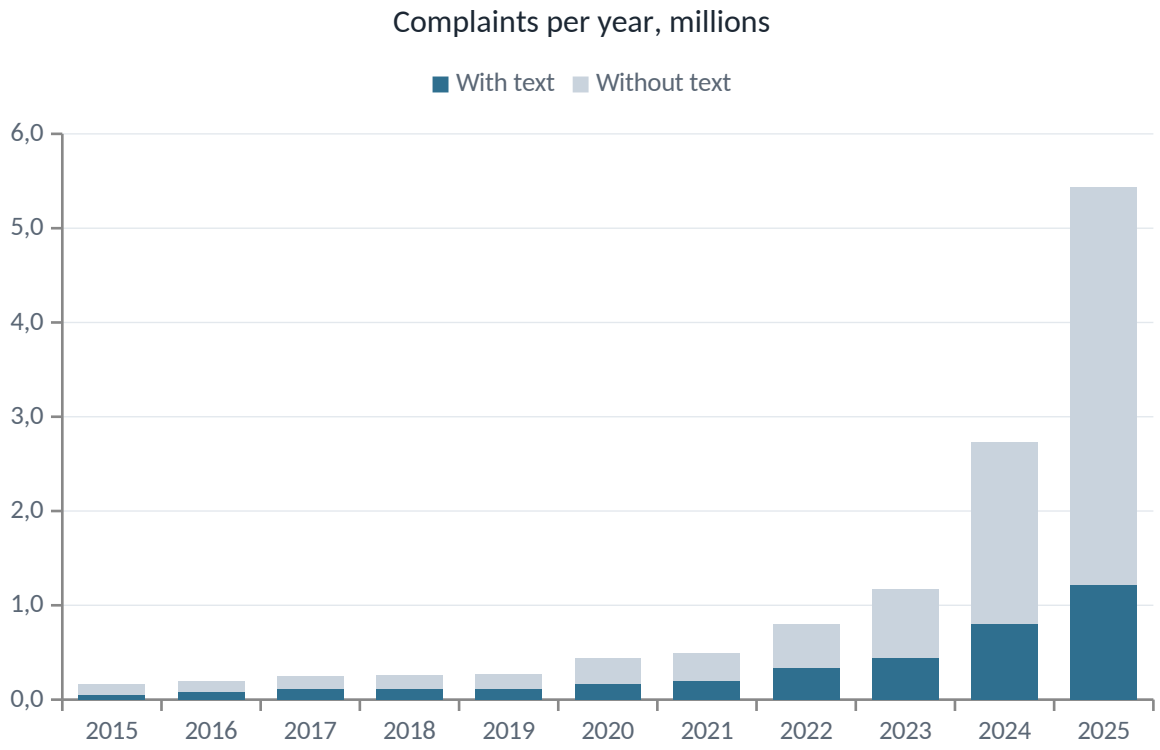
One raw row and the role of each field

Field	Value (complaint 11425997)	Role
Date received	2025-01-07	time split
Product	Mortgage	LABEL
Narrative	“Property Taxes in Escrow did not get paid to my village by the due date…”	MAIN FEATURE
Company · State · Channel	Shellpoint Partners, LLC · WI · Web	features
Date sent to company	2025-01-07	excluded: leakage
Company response	Closed with explanation	excluded: leakage
Complaint ID	11425997	join key

Data sample: one real complaint per class

Class (label)	Company	Complaint text
Checking or savings	KeyCorp	*I am disputing a negative account reported by KeyBank that is blocking me from opening any bank accounts…*
Credit or prepaid card	Synchrony	*I submitted proof of product return as requested. I received letters from Synchrony Bank…*
Credit reporting	TransUnion	*The information listed below, which appears on my credit report, is the result of that…*
Debt collection	Experian	*There are collection accounts on my report that I believe contain inaccurate information…*
Debt or credit mgmt	TransUnion	*I requested proof of late payments. I requested the late payments be investigated…*
Money transfer	Block, Inc.	*I contacted Cash app and they refuse to give my money back even after I showed proof it was a scam…*
Mortgage	Shellpoint	*I have paid more than XXXX of my mortgage… but the Newrez refused to cancel my PMI payment…*
Payday / personal loan	Regional Mgmt	*They changed the approved loan amount & terms 3 different times…*
Student loan	MOHELA	*Errors in loan balances, misapplied payments, and wrongful denials of repayment plans…*
Vehicle loan or lease	Ford Credit	*I set this account up on autopay… for some reason it was not set…*

3.8 million complaints with text, 2015–2025



16.6 M

complaints in the snapshot

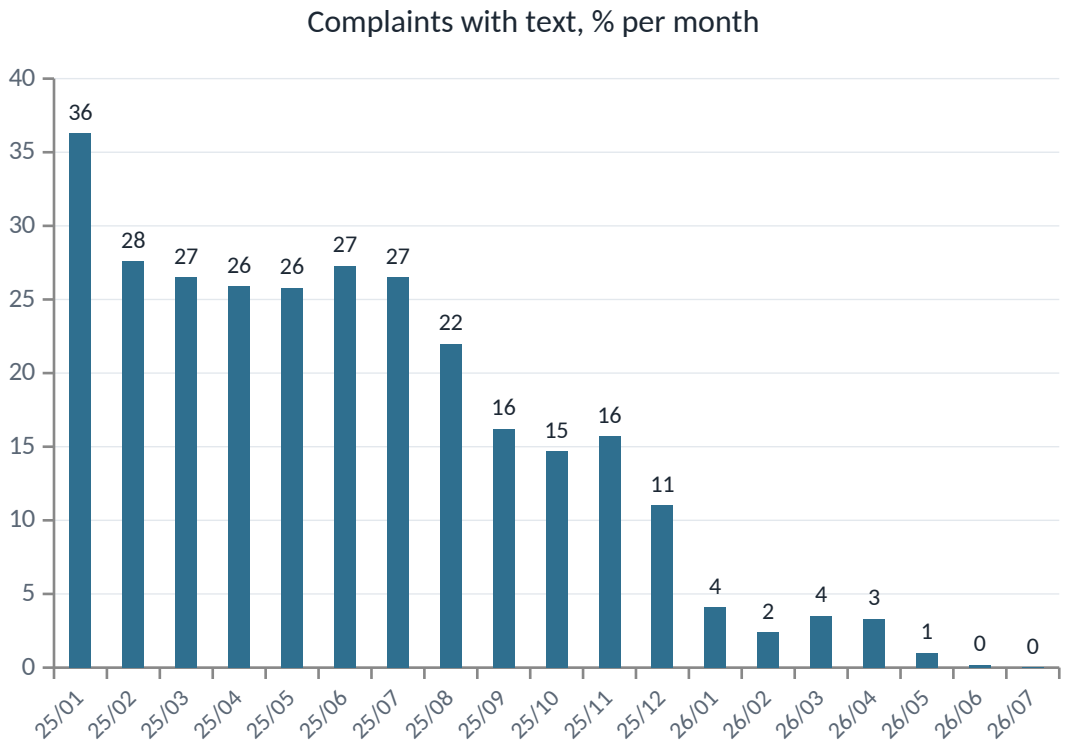
3.8 M

with text

1.5 M

in the training window

Finding: CFPB stopped publishing complaint text in 2026



26% → 0%

complaints with text,
2025-07 → 2026-07

Design response

- Snapshot text in DVC
- Replay one month per run
- Quality gate catches it

The label set changed twice; a versioned map absorbs it

2011 – 2017

- Credit reporting
- Bank account or service
- Consumer Loan

2017 – 2023

- Credit reporting, credit repair services…
- Credit card or prepaid card
- Checking or savings account

2023 – today

- Credit reporting or other personal consumer reports
- Debt or credit management (new)

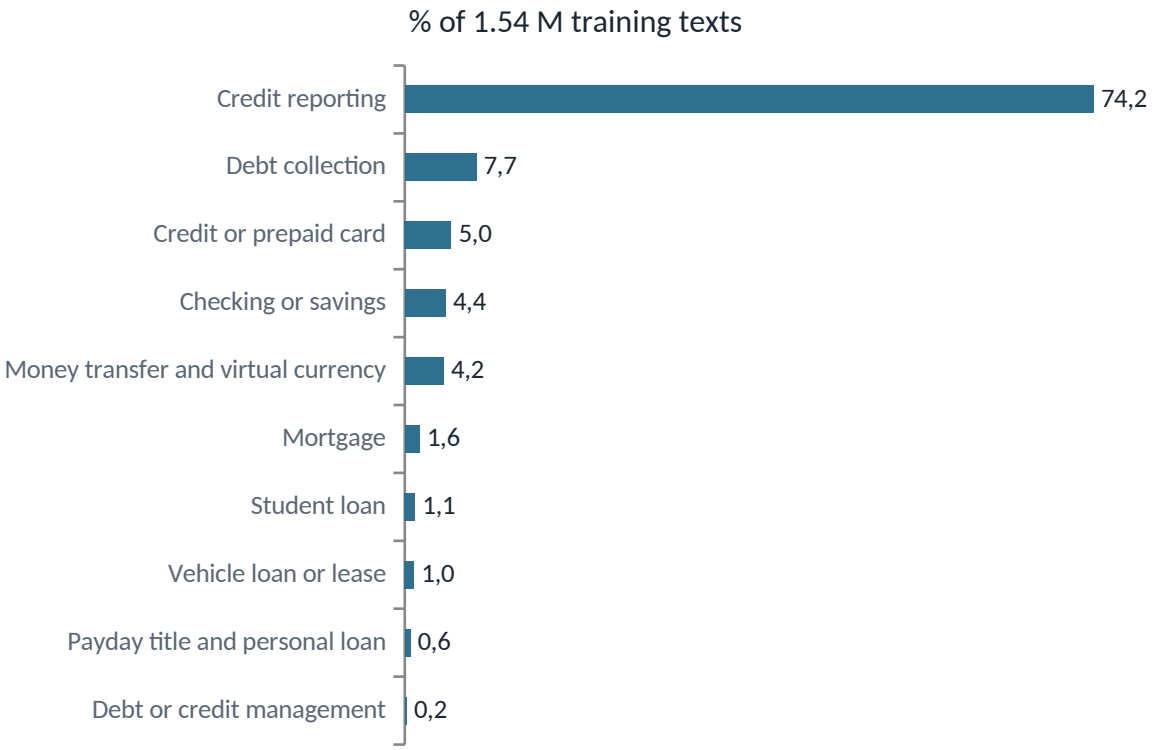
→

20 → 10

product names → stable classes

taxonomy_map.csv

Training texts are heavily imbalanced



Macro-F1

main metric

Class weights

in training

Recall guard

≤ 3 pp loss per class

Labels are noisy: real examples from the sample

Labelled: Debt collection

“My credit report shows collection accounts that I believe include inaccurate information…” (to TransUnion)

Labelled: Debt or credit management

“I requested proof of late payments… TransUnion has not sent proof…” (to TransUnion)

Both are really about a credit report.

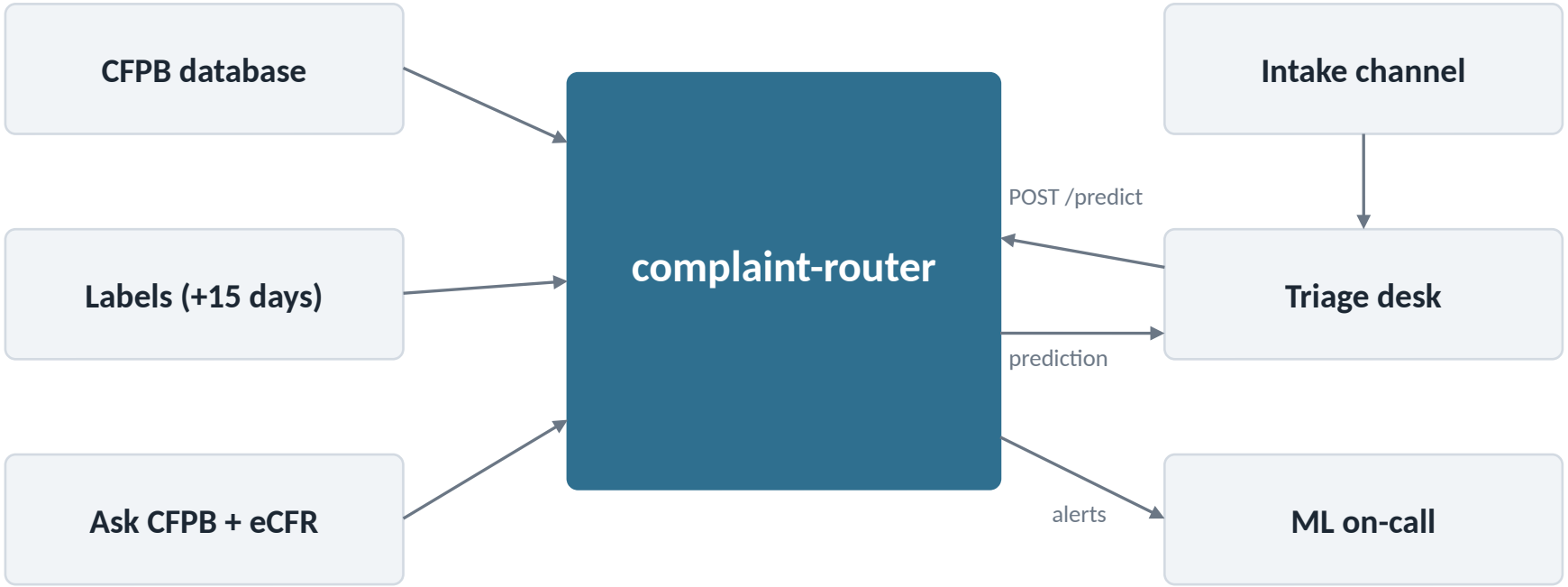
- Lab 2: hand-label 200 rows to measure noise
- Human overrides beat CFPB labels; model predictions are never labels

The data passes all four checks the labs require

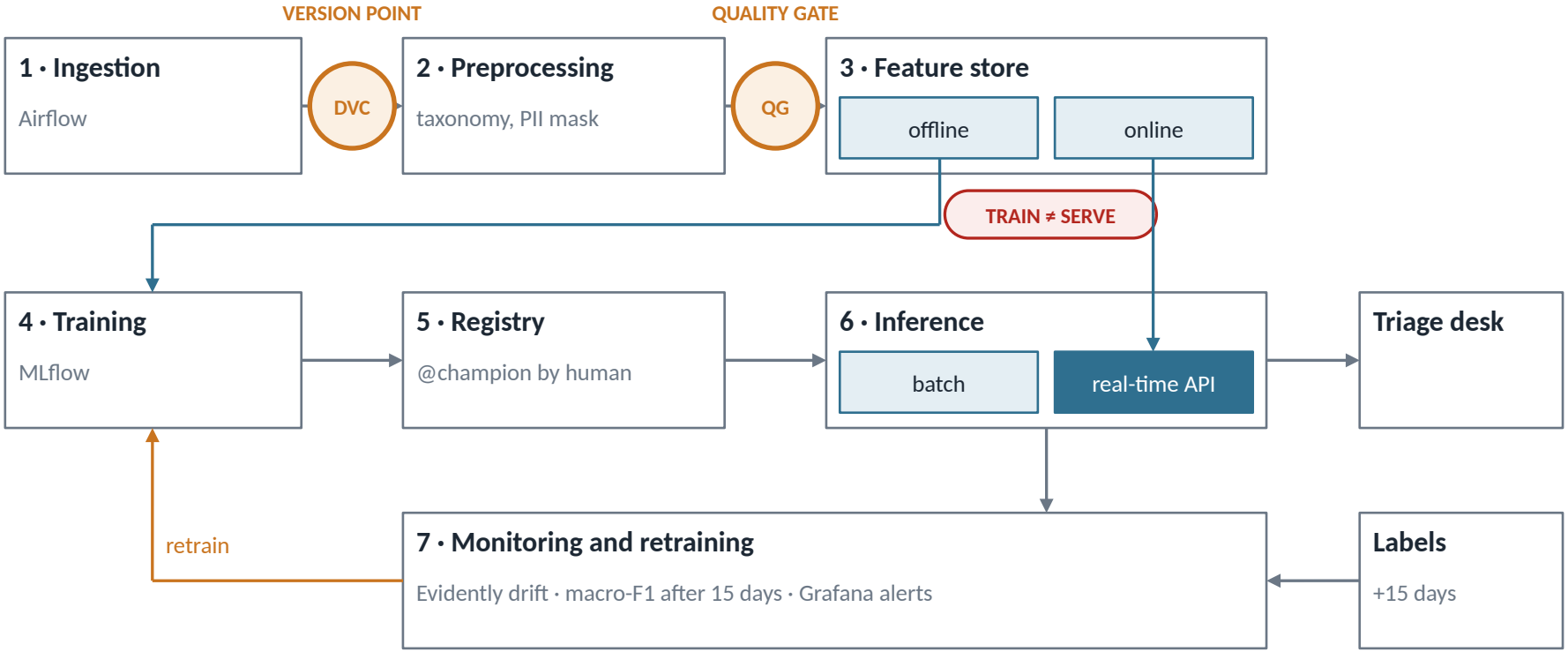
PASS	**Time axis**	daily dates since 2011
PASS	**History to hold out**	3.8 M texts; quarter-long val and test
PASS	**Label delay**	15 days (CFPB publication rule)
PASS	**Text corpus**	narratives + Ask CFPB + eCFR regulations



System context: one box, every external party around it



Data flow on the seven layers, with the three marks



Model: start from rules, keep the candidate simple

Baseline 0

majority class

Baseline 1

keyword rules

Candidate

TF-IDF + logistic regression, calibrated

Feature	Training	Serving
Complaint text	Parquet	request, same code
Channel, state	CFPB fields	request
Company	normalized	raw → alias table

Company is the one skew risk: monitored.

Monitoring: what we can see before the labels arrive



Days 0–14: proxies

- Prediction drift: PSI > 0.2
- New-word rate +50%
- Confidence drop > 0.05
- p95 > 300 ms or errors > 1%
- Stale data > 3 days

Day 15+: labels

- macro-F1 < 0.75 → alert + retrain
- Real misrouting rate

Runs on a laptop; every component has a priced cloud twin

Local	AWS	$/mo
MinIO	Amazon S3	3
PostgreSQL	Amazon RDS	15
Airflow	Amazon MWAA	250
MLflow	SageMaker MLflow	150
FastAPI on Minikube	Amazon EKS	136
Prometheus + Grafana	CloudWatch	15
pgvector (Lab 3)	OpenSearch	175
LLM (Lab 3)	Bedrock	160

$904

estimated per month

$800

ceiling

$729

with pgvector in RDS

Known risks, each with its answer

Text source stopped

DVC snapshot + replay

Taxonomy change

map + quality gate

Train/serve skew

shared preprocessing

15 blind days

proxy monitoring

4 GB laptop

linear model, hosted LLM

Feedback loop

labels never from predictions

This project carries all five labs

Lab	Needs	This project provides	Fit
1 · Design	design + data justification	design.md, data profile, sample, CI	Done
2 · Data + MLflow	batches, held-out period	monthly replay of 3.8 M texts	Fits
3 · LLM + retrieval	LLM baseline, corpus	zero-shot LLM vs TF-IDF; Ask CFPB + eCFR	Fits
4 · Serving + drift	servable model, real drift	FastAPI; 2023 taxonomy change, volume surge	Fits
5 · Agent	tasks to automate	drift review, model comparison, promotion note	Fits

Caveat: no new complaint text after 2025, so new batches are replayed history.

How the next four labs grow this system

Lab 2

Pipeline + MLflow

- Airflow DAG
- DVC + quality gate
- baselines in MLflow

Lab 3

LLM + retrieval

- LLM vs TF-IDF
- pgvector corpus
- cited rationales

Lab 4

Serving + drift

- API on Minikube
- Evidently + Grafana
- drift alert to human

Lab 5

Agent

- reads drift reports
- compares models
- drafts promotion

Where to find everything

Path	What
docs/design.md	design document
docs/data-profile.md	all data numbers
data/sample/complaints_sample.csv	200 real complaints
configs/taxonomy_map.csv	20 names → 10 classes
scripts/profile_data.py	reproduces the profile
Makefile · uv.lock · CI	make setup, make lint

Repository

github.com/
goodfrog33/
kse-mlops-
complaint-router

tag lab-1 · CI green